

Caleb Williams: Sack Avoidance Projection

A Bayesian Hierarchical AR(1) Model of Pressure-to-Sack Rate

Asa Arnold

Abstract

This paper estimates Caleb Williams' true sack avoidance skill and projects his 2026 pressure-to-sack rate (P2S) using a Bayesian hierarchical AR(1) model fit to 333 qualifying QB-seasons of PFF data (2018-2025). P2S rate, the fraction of defensive pressures that convert to sacks, isolates QB pocket awareness from offensive line quality. Williams posted a 28.2% P2S rate in his 2024 rookie season, ranking near the bottom of qualifying starters. In 2025, he improved to 9.6%, a historic 18.6 percentage point swing and 3.3 standard deviations below the mean year-over-year change. The model finds that P2S rate carries near-zero year-to-year autocorrelation once stable between-QB differences are controlled (lag coefficient near zero, $P(\text{beta} > 0) = 14\%$). Partial pooling shrinks Williams' estimated true talent toward the population mean; the posterior median settles at 13.2% (80% CrI: 11.0-15.5%). The central Year 3 projection is 13.0%, well below the league mean of 18.7% and indicating genuine elite-level sack avoidance skill. The model's scenario analysis finds that time-to-throw changes produce less than 0.5 percentage points of spread on the projection, meaning skill, not mechanics, drives the 2026 forecast.

1. Problem Framing

Sack rate is one of the most visible metrics attached to a quarterback's evaluation, yet it conflates two distinct phenomena: offensive line quality and quarterback decision-making. A QB behind a porous line will face more pressures regardless of skill, inflating his raw sack total. Pressure-to-sack rate (P2S) corrects for this by conditioning on pressures already generated, isolating the QB's contribution to the conversion rate.

Caleb Williams' two-year arc presents an unusual statistical challenge. His 28.2% P2S rate in 2024 placed him among the worst in the league. His 9.6% rate in 2025 placed him near the historic low. The 18.6 percentage point improvement is the largest year-over-year swing in the 2018-2025 dataset by a wide margin. The core question this analysis addresses: how much of that swing represents a durable change in skill versus regression to the mean?

With only two seasons of data, a naive point estimate of Williams' true talent is unreliable. A Bayesian framework is appropriate here because it quantifies uncertainty explicitly rather than collapsing it to a single number. Hierarchical pooling allows the model to borrow strength from the population of 100 qualifying QBs while still respecting Williams' individual data.

Table 1. Caleb Williams P2S Rate Summary

Season	P2S Rate	vs. League Mean	Percentile	Z-Score (YoY)
2024 (Y1)	28.2%	+9.5 pp	~86th	—

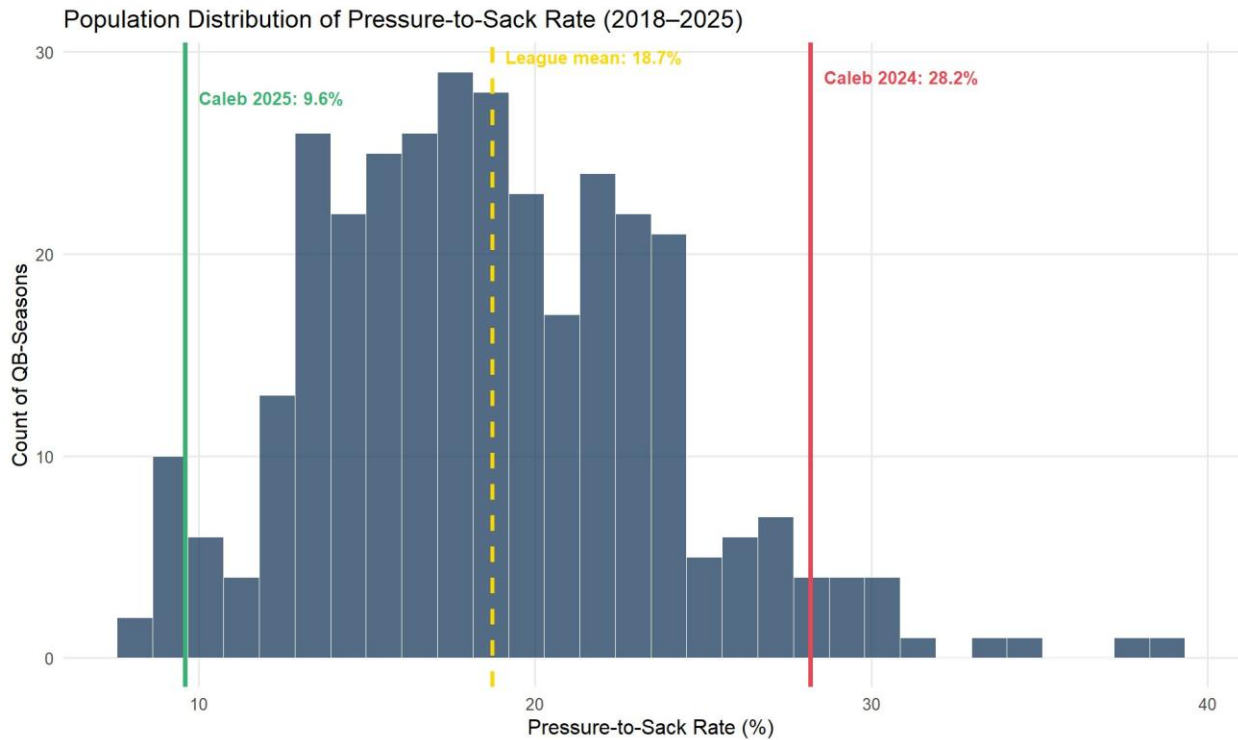
2025 (Y2)	9.6%	-9.1 pp	~5th	-3.3 SD
2026 Proj.	13.0%	-5.7 pp	~15th	—

Y1 and Y2 observed rates from PFF. Y3 projection = posterior predictive median (neutral time-to-throw scenario).

2. Data and Exploratory Analysis

The dataset draws on PFF passing summary files for the 2018-2025 NFL seasons. QB-seasons are included if the player logged at least 150 dropbacks (to establish a stable rate) and at least 400 passing snaps (to qualify as a genuine starter). This yields 333 raw QB-seasons across 8 seasons, reduced to 231 modeling observations after each QB's first qualifying season is reserved as the AR(1) lag for their second.

Figure 1. Population Distribution of Pressure-to-Sack Rate (2018-2025)



The league-wide distribution of P2S rate across all qualifying QB-seasons. Williams' 2025 rate (green) sits near the extreme left tail; his 2024 rate (red) sits in the right tail. The yellow dashed line marks the population mean of 18.7%.

Figure 1 establishes the population context. The distribution is roughly bell-shaped with a mean near 18.7% and a standard deviation of approximately 5.5 percentage points. Williams' 2025 rate of 9.6% and his 2024 rate of 28.2% are both notable extremes in opposite directions, which motivates a careful assessment of signal versus noise.

Figure 2. Lag-1 Persistence of QB Pressure-to-Sack Rate

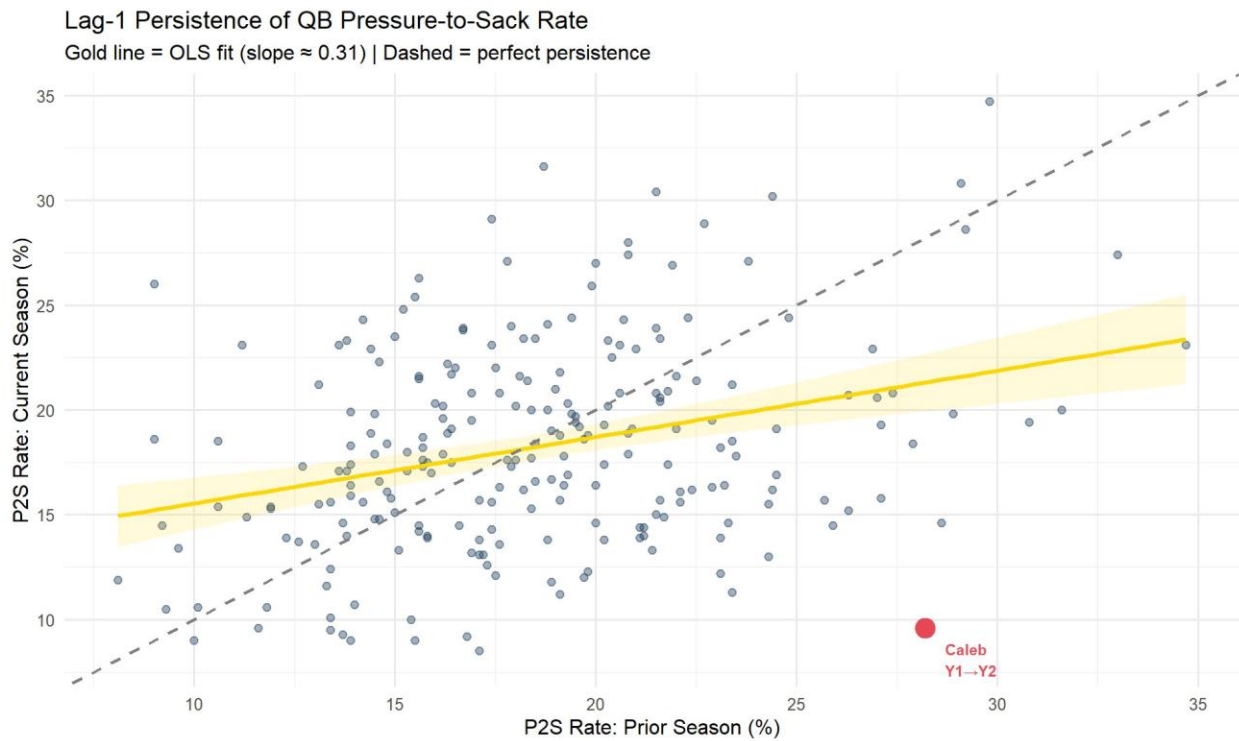


Figure 2 reveals the persistence structure of the metric. The OLS slope of approximately 0.31 means that a QB one standard deviation above the mean in year T is expected to be only 0.31 standard deviations above in year T+1. This low persistence is what makes Williams' improvement so statistically unusual and motivates heavy shrinkage in the Bayesian model.

3. Methods

The model treats each QB-season as a binomial draw: sack counts are modeled as $\text{Binomial}(n, p)$, where n is the number of pressures and p is the latent conversion probability. A player-level random intercept captures stable between-QB differences in true sack avoidance skill. The fixed effects include the prior-season P2S rate (AR(1) lag), the change in average time-to-throw from the prior season, the raw average time-to-throw, scramble rate, career year, and starter year.

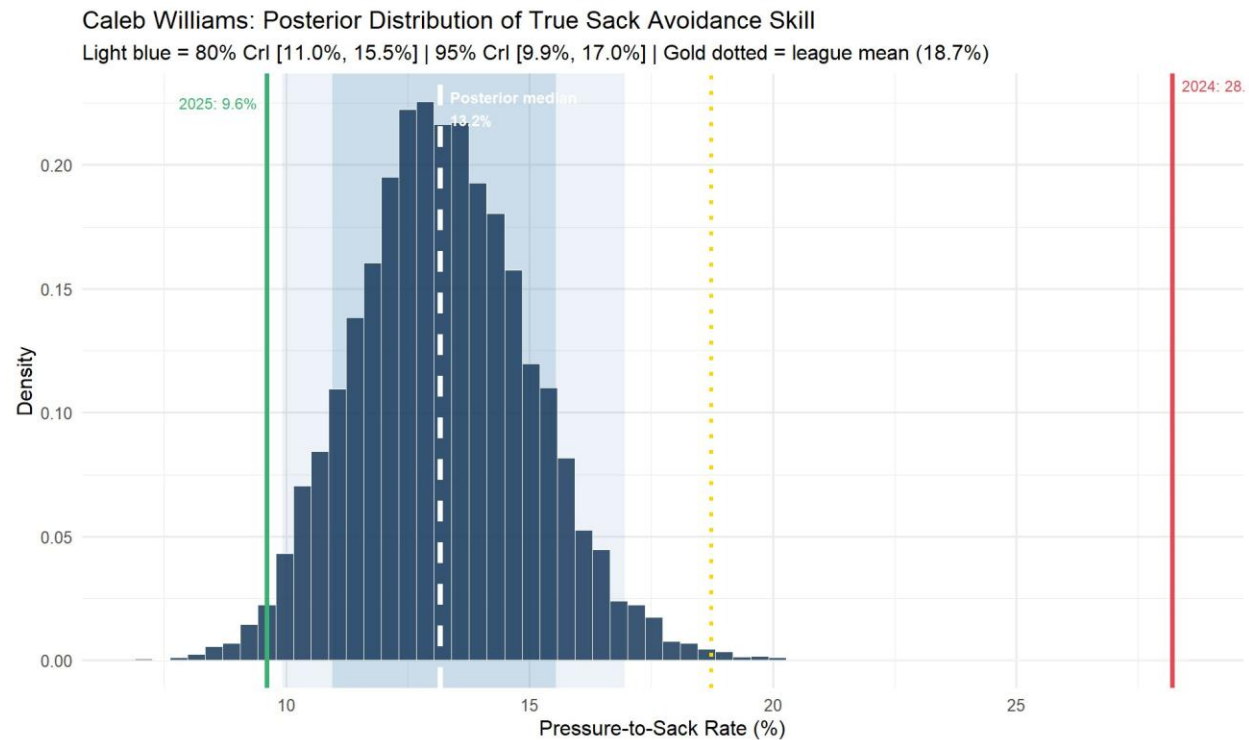
Formally: $\text{logit}(p_{it}) = \alpha_{\text{QB}_i} + \beta_{\text{lag}} * P2S(t-1) + \beta_{\text{tth}} * TTT + \beta_{\text{scramble}} * SCR + \beta_{\text{career}} * CYR + \epsilon$. Prior distributions were set conservatively: $\text{Normal}(-1.5, 0.3)$ on the intercept, $\text{Normal}(0, 0.5)$ on fixed effects, and $\text{Normal}(0, 0.2)$ on the player-level SD.

Sampling used brms with the CmdStan backend: 4 chains, 3,000 iterations with 1,000 warmup, $\text{adapt_delta} = 0.95$. Convergence was confirmed by R-hat values below 1.01 and effective sample size ratios above 0.1 for all parameters.

Williams' Year 3 projection is generated from the posterior predictive distribution using his Year 2 observed covariates, a career year of 3, and three time-to-throw scenarios (neutral, +0.15s regression, -0.15s improvement). Because the lag coefficient is near zero, the projection is driven almost entirely by his estimated player random intercept.

4. Results

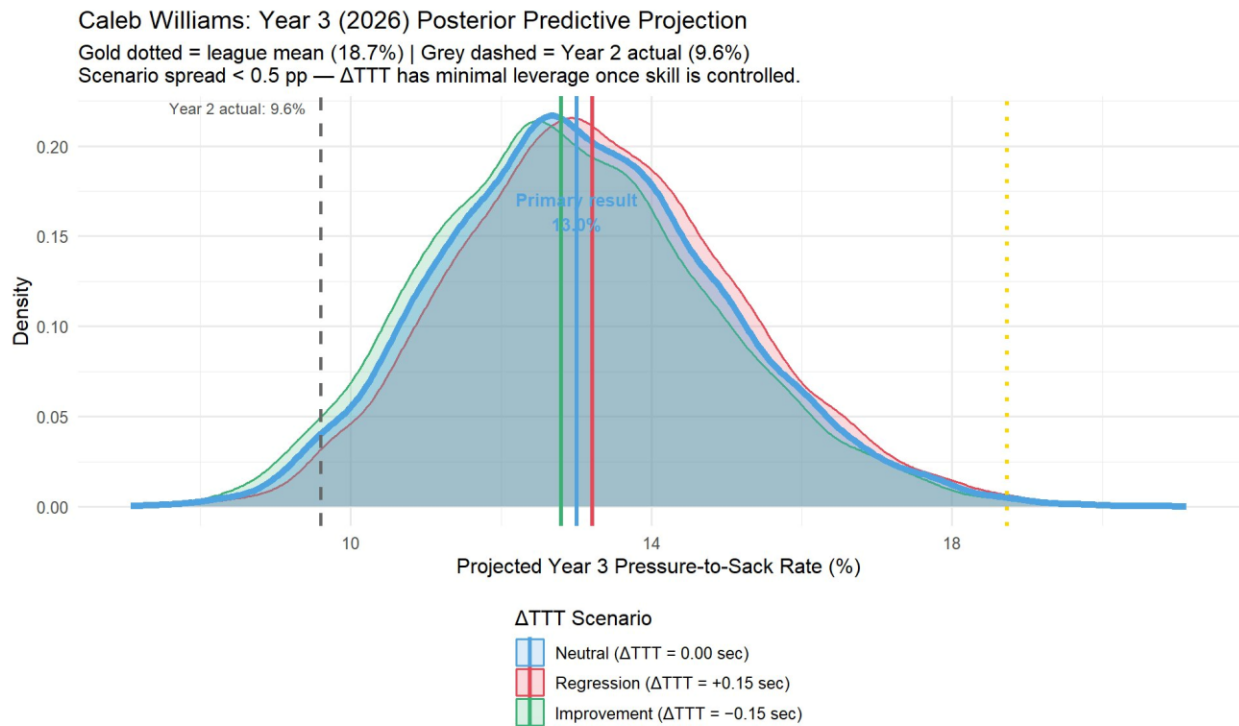
Figure 3. Posterior Distribution of Williams' True Sack Avoidance Skill



The posterior distribution of Caleb Williams' estimated true P2S rate after partial pooling. The posterior median is 13.2% (80% CrI: 11.0-15.5%, 95% CrI: 9.9-17.0%). The league mean (18.7%, gold dotted) lies well to the right of the bulk of the posterior mass.

Figure 3 shows the model's best estimate of Williams' underlying sack avoidance talent. The key result is that the posterior median of 13.2% is substantially below the league mean of 18.7%, but also above his 2025 observed rate of 9.6%. This is partial pooling in action: the model treats his 9.6% as a mixture of real skill and noise, settling on an estimate that is meaningfully elite but not as extreme as a single season would suggest.

Figure 4. Year 3 (2026) Posterior Predictive Projection



The Year 3 posterior predictive distribution under three time-to-throw scenarios. All three scenarios produce nearly identical distributions, with medians ranging from 12.8% to 13.2%. The scenario spread of less than 0.5 pp confirms that Williams' estimated skill, not mechanics, drives the forecast.

Figure 4 presents the 2026 projection. The central estimate is 13.0% under the neutral scenario (no change in time-to-throw), with a wide posterior standard deviation of approximately 3.0 percentage points reflecting the limited individual data. The three scenarios produce medians that differ by less than half a percentage point, confirming that the AR(1) lag coefficient's near-zero value makes time-to-throw adjustments largely irrelevant to the forecast.

At an assumed pressure volume of 240, a 13% P2S rate projects to approximately 31 sacks. This compares favorably to his Year 1 pace and is consistent with a quarterback operating at a genuine elite level of pocket escapability.

5. Implications

The primary takeaway for the Bears organization is that Williams' 2025 improvement appears to be substantially real. The model is skeptical of his 9.6% observed rate as a literal true-talent estimate, but even after heavy shrinkage toward the 100-QB population mean, his posterior median of 13.2% ranks in approximately the 15th percentile of the qualifying QB distribution. For a player in his third year, that represents a genuine competitive advantage.

A secondary finding with broader implications: P2S rate has near-zero year-to-year persistence once stable skill differences are controlled. This means that a QB's prior-season rate adds almost no marginal predictive value when his individual intercept is already estimated. Organizations relying on raw year-over-year P2S changes to evaluate QB development are measuring mostly noise.

The model also underscores the limits of two seasons of data. Williams' random intercept is currently identified by one lag-to-current transition and the population prior. The 2026 season will be the first in

which his individual estimate is meaningfully constrained by his own three-year history. Projections generated after Year 3 will carry substantially narrower uncertainty intervals.

Limitations

P2S rate controls for OL quality better than raw sack count, but OL quality still affects pressure type. A strong line may generate cleaner, more escapable pressures; a weak line may yield stunt-heavy designs that are harder to escape regardless of QB skill. Play-level data would be required to remove this residual confound.

Pressure volume in 2026 is assumed, not modeled. Projecting sack counts requires an assumption about how many pressures Williams will face, which depends on OL performance, opposing defensive schemes, and game script. Results are therefore presented across a 200-280 pressure range rather than as a single forecast.

Finally, the PFF pressure designation is subjective and assessor-dependent. Systematic differences in how pressures are coded across seasons could introduce noise into the P2S metric that the model cannot separate from true variation in QB performance.
